

Exercise Sheet 4 — Solutions

Exercise 1 – Planes and directions in crystals

(a) Miller indices in FCC: conventional vs. primitive cell

For an fcc crystal, the conventional cubic unit cell has lattice vectors:

$$\vec{a}_x = a\hat{x}, \quad \vec{a}_y = a\hat{y}, \quad \vec{a}_z = a\hat{z}.$$

Its primitive vectors (with one lattice point per cell) are commonly chosen as:

$$\vec{a}_1 = \frac{a}{2}(\hat{y} + \hat{z}), \quad \vec{a}_2 = \frac{a}{2}(\hat{z} + \hat{x}), \quad \vec{a}_3 = \frac{a}{2}(\hat{x} + \hat{y}).$$

Miller indices (hkl) refer to a family of lattice planes with normal vector proportional to

$$\vec{n} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3,$$

where $\vec{b}_i$ are reciprocal lattice vectors, satisfying $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$.

Conventional cell planes:

- (100): Planes intersect $x = a, y \rightarrow \infty, z \rightarrow \infty$ => spacing $d = a$. These are the planes perpendicular to x , cutting the cube at $x = na$ ($n \in \mathbb{Z}$). In FCC, atoms lie at corners and face centers; the (100) planes pass through face centers of the yz -faces.
- (110): Intercepts (a, a, ∞) => planes diagonal in xy , perpendicular to $[110]$. Spacing $d_{110} = a/\sqrt{2}$. In FCC, the (110) planes are densely packed and contain rows of atoms along $[001]$.
- (111): Intercepts (a, a, a) => planes normal to $[111]$, the close-packed planes in FCC. Spacing $d_{111} = a/\sqrt{3}$. Each (111) plane contains a triangular lattice of atoms.

Primitive cell Miller indices: To obtain Miller indices *with respect to the primitive cell*, we must express the *normal vector* to the plane in the basis of the *primitive reciprocal lattice vectors*.

First compute the reciprocal lattice vectors of the primitive fcc cell.

Let $\vec{a}_1 = \frac{a}{2}(\hat{y} + \hat{z})$, etc., then the volume of the primitive cell is:

$$V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = \frac{a^3}{4}.$$

Standard formula for reciprocal vectors:

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{V}, \quad \text{etc.}$$

Compute:

$$\vec{a}_2 \times \vec{a}_3 = \left(\frac{a}{2}(\hat{z} + \hat{x})\right) \times \left(\frac{a}{2}(\hat{x} + \hat{y})\right) = \frac{a^2}{4}[(\hat{z} + \hat{x}) \times (\hat{x} + \hat{y})].$$

Expanding:

$$= \frac{a^2}{4}[\hat{z} \times \hat{x} + \hat{z} \times \hat{y} + \hat{x} \times \hat{x} + \hat{x} \times \hat{y}] = \frac{a^2}{4}[\hat{y} - \hat{x} + 0 + \hat{z}] = \frac{a^2}{4}(-\hat{x} + \hat{y} + \hat{z}).$$

Then:

$$\vec{b}_1 = 2\pi \frac{\frac{a^2}{4}(-\hat{x} + \hat{y} + \hat{z})}{a^3/4} = \frac{2\pi}{a}(-\hat{x} + \hat{y} + \hat{z}).$$

Similarly, by cyclic permutation:

$$\vec{b}_2 = \frac{2\pi}{a}(\hat{x} - \hat{y} + \hat{z}), \quad \vec{b}_3 = \frac{2\pi}{a}(\hat{x} + \hat{y} - \hat{z}).$$

Now, a plane with Miller indices $(hkl)_{\text{conv}}$ in the *conventional* cell has normal vector proportional to $h\hat{x} + k\hat{y} + l\hat{z}$. In reciprocal space, the corresponding $\vec{G}$ vector is:

$$\vec{G}_{\text{conv}} = \frac{2\pi}{a}(h\hat{x} + k\hat{y} + l\hat{z}).$$

We now express this $\vec{G}$ in the basis $(\vec{b}_1, \vec{b}_2, \vec{b}_3)$ to find the primitive-cell Miller indices $(h'k'l')$ such that:

$$\vec{G} = h'\vec{b}_1 + k'\vec{b}_2 + l'\vec{b}_3.$$

Write matrix form. Define matrix $B = [\vec{b}_1 \ \vec{b}_2 \ \vec{b}_3]$ (as columns):

$$B = \frac{2\pi}{a} \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}.$$

Let $\vec{G}_{\text{conv}} = \frac{2\pi}{a}(h, k, l)^T$. Then:

$$(h', k', l')^T = B^{-1} \vec{G}_{\text{conv}} = \frac{a}{2\pi} B^{-1} \cdot \frac{2\pi}{a}(h, k, l)^T = B^{-1}(h, k, l)^T.$$

Since $B = \frac{2\pi}{a}M$, with M the integer matrix above, then $B^{-1} = \frac{a}{2\pi}M^{-1}$. But since we only care about *direction* and *ratio* (Miller indices defined up to a scalar), we can invert M :

$$M = \begin{pmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{pmatrix}, \quad \det(M) = -4.$$

Adjugate:

$$M^{-1} = -\frac{1}{4} \begin{pmatrix} 0 & 2 & 2 \\ 2 & 0 & 2 \\ 2 & 2 & 0 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

Thus:

$$(h', k', l')^T = M^{-1}(h, k, l)^T = -\frac{1}{2} \begin{pmatrix} k+l \\ h+l \\ h+k \end{pmatrix}.$$

Multiply by -2 to get smallest integers (Miller indices are defined up to integer scale):

$$(h'k'l') = (k+l, h+l, h+k).$$

Now apply:

$$(100)_{\text{conv}}: h=1, k=0, l=0 \Rightarrow (0+0, 1+0, 1+0) = (0, 1, 1).$$

$$(110)_{\text{conv}}: h=1, k=1, l=0 \Rightarrow (1+0, 1+0, 1+1) = (1, 1, 2).$$

$$(111)_{\text{conv}}: h=1, k=1, l=1 \Rightarrow (1+1, 1+1, 1+1) = (2, 2, 2) \sim (1, 1, 1).$$

So the Miller indices *with respect to the primitive cell* are:

$(100)_{\text{conv}} \rightarrow (011)_{\text{prim}}$
$(110)_{\text{conv}} \rightarrow (112)_{\text{prim}}$
$(111)_{\text{conv}} \rightarrow (111)_{\text{prim}}$

(Note: (011) is equivalent to (110) in primitive indexing up to permutation—depending on labeling, but the formula above is consistent with chosen $\vec{a}_i$.)

Sketches (described) :

- (100): planes parallel to yz -faces, spaced by a . - (110): diagonal in xy , vertical, passing through edges and face centers. - (111): close-packed planes cutting cube corners; hexagonal atomic arrangement.

(b) Simple cubic: (110) plane spacing and reciprocal vector

For a simple cubic lattice with lattice constant a , conventional = primitive cell.

The interplanar spacing for Miller indices (hkl) is:

$$d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}.$$

Thus for (110):

$d_{110} = \frac{a}{\sqrt{1^2 + 1^2 + 0^2}} = \frac{a}{\sqrt{2}}.$
--

The reciprocal lattice of a simple cubic is also simple cubic, with primitive reciprocal vectors:

$$\vec{b}_1 = \frac{2\pi}{a}\hat{x}, \quad \vec{b}_2 = \frac{2\pi}{a}\hat{y}, \quad \vec{b}_3 = \frac{2\pi}{a}\hat{z}.$$

Hence the reciprocal lattice vector for family (hkl) is:

$$\vec{G}_{hkl} = h\vec{b}_1 + k\vec{b}_2 + l\vec{b}_3 = \frac{2\pi}{a}(h\hat{x} + k\hat{y} + l\hat{z}).$$

For (110) :

$$\vec{G}_{110} = \frac{2\pi}{a}(\hat{x} + \hat{y}).$$

This is parallel to the *normal* of the (110) planes in direct space — indeed, in cubic systems, the vector $[hkl]$ (direction indices) is parallel to the plane normal of (hkl) . So:

$$\vec{G}_{110} \parallel \hat{n}_{(110)} = \frac{1}{\sqrt{2}}(\hat{x} + \hat{y}).$$

Moreover, Bragg's law relates:

$$|\vec{G}_{hkl}| = \frac{2\pi}{d_{hkl}}.$$

Indeed,

$$|\vec{G}_{110}| = \frac{2\pi}{a}\sqrt{2} = \frac{2\pi}{a/\sqrt{2}} = \frac{2\pi}{d_{110}},$$

as expected.

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Indeed,

$$|\vec{G}_{110}| = \frac{2\pi}{a}\sqrt{2} = \frac{2\pi}{a/\sqrt{2}} = \frac{2\pi}{d_{110}},$$

as expected.

Orientation relative to planes: $\vec{G}_{110}$ is perpendicular to the (110) planes — by definition, $\vec{G}_{hkl}$ is normal to the (hkl) family. In diffraction, constructive interference occurs when scattering vector $\vec{q} = \vec{k}' - \vec{k} = \vec{G}_{hkl}$, i.e., when $\vec{q}$ is normal to the planes.

Exercise 2 – Diffraction and interference in 1D gratings

Consider N identical slits at positions $x_n = na$, $n = 0, \dots, N-1$.

The complex amplitude at far-field angle θ (small-angle approx: $k_x = k \sin \theta \approx k\theta$) is proportional to:

$$A_N(\theta) \propto \underbrace{\left[\sum_{n=0}^{N-1} e^{in ka \sin \theta} \right]}_{\text{interference}} \times \underbrace{A_{\text{single}}(\theta)}_{\text{single-slit diffraction}}.$$

For negligible slit width, $A_{\text{single}} \approx \text{const.}$, and

$$A_N \propto \frac{\sin\left(N\pi \frac{a \sin \theta}{\lambda}\right)}{\sin\left(\pi \frac{a \sin \theta}{\lambda}\right)} \equiv \frac{\sin(N\phi/2)}{\sin(\phi/2)}, \quad \phi = \frac{2\pi a \sin \theta}{\lambda}.$$

(a) $N = \infty$, negligible slit width

In the limit $N \rightarrow \infty$, the interference function becomes a **Dirac comb** (series of δ -functions) at:

$$a \sin \theta = m\lambda \quad \Leftrightarrow \quad \sin \theta_m = m \frac{\lambda}{a}, \quad m \in \mathbb{Z}.$$

So the diffraction pattern consists of **infinitely sharp, equally spaced Bragg peaks** (principal maxima) at angles satisfying the grating equation. Intensity $I \propto |A_N|^2$ diverges at these points (idealized as δ -functions). This is the **Fourier series** of a periodic train of δ -functions.

Physically: infinite periodicity $\Rightarrow$ perfectly sharp reciprocal lattice points.

(b) Finite slit width $w > 0$

Now the single-slit diffraction function modulates the interference envelope:

$$A_{\text{single}}(\theta) \propto \frac{\sin\left(\pi \frac{w \sin \theta}{\lambda}\right)}{\pi \frac{w \sin \theta}{\lambda}} = \text{sinc}\left(\frac{w \sin \theta}{\lambda}\right).$$

So total intensity:

$$I(\theta) \propto \left| \frac{\sin(N\phi/2)}{\sin(\phi/2)} \right|^2 \cdot \text{sinc}^2\left(\frac{w \sin \theta}{\lambda}\right).$$

Effect:

Principal maxima still at $\sin \theta = m\lambda/a$,

But overall intensity envelope has minima at $\sin \theta = n\lambda/w$, $n = \pm 1, \pm 2, \dots$

If $w = a$, envelope first zero at $\theta = \lambda/a$, i.e., coincides with $m = 1$ peak $\Rightarrow$ first-order diffraction suppressed.

If $w < a$, more orders visible.

Thus, increasing slit width narrows the diffraction envelope and may suppress higher-order peaks.

(c) Finite N : 20, 4, 2 slits (negligible width)

For finite N , the sharp δ -peaks broaden into peaks of finite width.

The angular width (full-width at first zero) of the m -th principal maximum is:

$$\Delta(\sin \theta) \approx \frac{2\lambda}{Na}.$$

- As N decreases:

Peak width $\propto 1/N \Rightarrow$ broader peaks.

Number of secondary maxima between principal peaks = $N - 2$.

Peak intensity $I_{\text{max}} \propto N^2$.

Specifically:

$N = 20$: 18 secondary maxima between principal peaks. Peaks still narrow and sharp; good resolution.

$N = 4$: 2 secondary maxima. Peaks significantly broadened. Intensity ratio principal/secondary = $(4/2)^2 = 4$.

$N = 2$: **double-slit interference**. No secondary maxima — only $\cos^2$ fringe pattern:

$$I(\theta) \propto \cos^2\left(\frac{\pi a \sin \theta}{\lambda}\right).$$

First zeros at $\sin \theta = \pm\lambda/(2a)$, principal maxima at $m\lambda/a$.

Thus, the transition from crystal ($N \sim 10^{10}$) to molecule ($N = 2$) drastically reduces resolution and eliminates satellite peaks.

Exercise 3 – Reciprocal lattice of a 2D Bravais lattice

Given:

$$\vec{a}_1 = (0.4, 0) \text{ nm}, \quad \vec{a}_2 = (0.1, 0.2) \text{ nm}.$$

(a) Area and A_c

In 2D, $\vec{a}_1 \times \vec{a}_2 = (a_{1x}a_{2y} - a_{1y}a_{2x})\hat{z}$. So:

$$A = (\vec{a}_1 \times \vec{a}_2) \cdot \hat{e}_z = (0.4)(0.2) - (0)(0.1) = 0.08 \text{ nm}^2.$$

Thus:

$$A = 0.08 \text{ nm}^2, \quad A_c = A^2 = 6.4 \times 10^{-3} \text{ nm}^4.$$

(b) Reciprocal vectors $\vec{b}_1, \vec{b}_2$

In 2D, the reciprocal vectors satisfying $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ are:

$$\vec{b}_1 = 2\pi \frac{\hat{z} \times \vec{a}_2}{A}, \quad \vec{b}_2 = 2\pi \frac{\vec{a}_1 \times \hat{z}}{A},$$

since $\hat{z} \times \vec{a} = (-a_y, a_x)$.

Compute:

$$\hat{z} \times \vec{a}_2 = (-a_{2y}, a_{2x}) = (-0.2, 0.1) \text{ nm},$$

so:

$$\vec{b}_1 = 2\pi \frac{(-0.2, 0.1)}{0.08} = 2\pi(-2.5, 1.25) \text{ nm}^{-1} = (-5\pi, 2.5\pi) \text{ nm}^{-1} \approx (-15.708, 7.854) \text{ nm}^{-1}.$$

Similarly:

$$\vec{a}_1 \times \hat{z} = (a_{1y}, -a_{1x}) = (0, -0.4),$$

so:

$$\vec{b}_2 = 2\pi \frac{(0, -0.4)}{0.08} = 2\pi(0, -5) = (0, -10\pi) \text{ nm}^{-1} \approx (0, -31.416) \text{ nm}^{-1}.$$

Magnitudes:

$$|\vec{b}_1| = \sqrt{(-5\pi)^2 + (2.5\pi)^2} = \pi\sqrt{25 + 6.25} = \pi\sqrt{31.25} = \pi \cdot 5\sqrt{1.25} = \frac{5\pi\sqrt{5}}{2} \approx 17.56 \text{ nm}^{-1}.$$

$$|\vec{b}_2| = 10\pi \approx 31.42 \text{ nm}^{-1}.$$

(c) Verification

Compute dot products:

$$\vec{a}_1 \cdot \vec{b}_1 = (0.4, 0) \cdot (-5\pi, 2.5\pi) = 0.4 \cdot (-5\pi) = -2\pi.$$

Wait — sign issue! Let's re-check orientation.

The correct 2D reciprocal definition preserving $\vec{a}_i \cdot \vec{b}_j = 2\pi\delta_{ij}$ is:

$$\vec{b}_1 = 2\pi \frac{R_{90}\vec{a}_2}{A}, \quad \vec{b}_2 = 2\pi \frac{R_{-90}\vec{a}_1}{A},$$

where $R_{90} = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ rotates $\vec{v} \rightarrow \hat{z} \times \vec{v}$.

But $R_{90}\vec{a}_2 = (-a_{2y}, a_{2x}) = (-0.2, 0.1)$ — correct.

Yet $\vec{a}_1 \cdot \vec{b}_1 = -2\pi$, not $+2\pi$.

This suggests the primitive cell is defined with *negative* orientation (since $A = +0.08 > 0$, but our cross product used $\hat{z} = \vec{a}_1 \times \vec{a}_2 / A$, so actually $\vec{a}_1 \times \vec{a}_2 = +0.08\hat{z}$, meaning right-handed).

Let's compute explicitly:

$$\vec{a}_1 = (0.4, 0), \quad \vec{a}_2 = (0.1, 0.2).$$

We want $\vec{b}_1$ such that:

$$\vec{a}_1 \cdot \vec{b}_1 = 2\pi, \quad \vec{a}_2 \cdot \vec{b}_1 = 0.$$

Let $\vec{b}_1 = (u, v)$. Then:

$$0.4u = 2\pi \Rightarrow u = 5\pi, \quad 0.1u + 0.2v = 0 \Rightarrow 0.1(5\pi) + 0.2v = 0 \Rightarrow 0.5\pi + 0.2v = 0 \Rightarrow v = -2.5\pi.$$

So:

$$\vec{b}_1 = (5\pi, -2.5\pi) \text{ nm}^{-1} \approx (15.708, -7.854) \text{ nm}^{-1}.$$

Similarly, $\vec{b}_2 = (p, q)$ satisfying:

$$\vec{a}_1 \cdot \vec{b}_2 = 0 \Rightarrow 0.4p = 0 \Rightarrow p = 0, \vec{a}_2 \cdot \vec{b}_2 = 2\pi \Rightarrow 0.1 \cdot 0 + 0.2q = 2\pi \Rightarrow q = 10\pi.$$

So:

$$\boxed{\vec{b}_2 = (0, 10\pi) \text{ nm}^{-1} \approx (0, 31.416) \text{ nm}^{-1}}.$$

Now verify:

$$\vec{a}_1 \cdot \vec{b}_1 = 0.4 \cdot 5\pi = 2\pi \quad \checkmark \quad \vec{a}_2 \cdot \vec{b}_1 = 0.1 \cdot 5\pi + 0.2 \cdot (-2.5\pi) = 0.5\pi - 0.5\pi = 0 \quad \checkmark \quad \vec{a}_1 \cdot \vec{b}_2 = 0.4 \cdot 0 = 0 \quad \checkmark \quad \vec{a}_2 \cdot \vec{b}_2 = 0.1 \cdot 0 + 0.2 \cdot 10\pi = 2\pi$$

Magnitudes (corrected):

$$|\vec{b}_1| = \sqrt{(5\pi)^2 + (-2.5\pi)^2} = \pi\sqrt{25 + 6.25} = \pi\sqrt{31.25} = \frac{5\pi\sqrt{5}}{2} \approx 17.56 \text{ nm}^{-1}, \quad |\vec{b}_2| = 10\pi \approx 31.42 \text{ nm}^{-1}.$$

(d) Reciprocal lattice sketch and 1st Brillouin zone

Reciprocal lattice points: $\vec{G}_{hk} = h\vec{b}_1 + k\vec{b}_2$, $h, k \in \mathbb{Z}$.

- $\vec{b}_1 \approx (15.7, -7.85)$, $\vec{b}_2 \approx (0, 31.42)$. - So points lie on a lattice skewed downward: for fixed k , h shifts right and down.

First Brillouin zone = Wigner-Seitz cell around $\Gamma = (0, 0)$:

Construct perpendicular bisectors between Γ and nearest reciprocal points.

Nearest $\vec{G}$:

$$\vec{G}_{10} = \vec{b}_1 \approx (15.7, -7.85), \quad \vec{G}_{\bar{1}0} = -\vec{b}_1, \quad \vec{G}_{01} = (0, 31.42), \quad \vec{G}_{0\bar{1}} = (0, -31.42).$$

But also $\vec{G}_{1\bar{1}} = \vec{b}_1 - \vec{b}_2 = (15.7, -39.3)$ — farther.

Compute distances:

$$|\vec{b}_1| \approx 17.56, \quad |\vec{b}_2| = 31.42, \quad |\vec{b}_1 - \vec{b}_2| = \sqrt{15.7^2 + (-39.3)^2} \approx 42.3.$$

So nearest neighbors: $\pm\vec{b}_1$, and possibly $\pm(\vec{b}_2 - \vec{b}_1)$? Check $|\vec{b}_2 - \vec{b}_1| = |\vec{b}_1 - \vec{b}_2|$ — same 42.3 $\Rightarrow$ not nearest.

Actually, also $\vec{G}_{11} = \vec{b}_1 + \vec{b}_2 = (15.7, 23.57)$, $|\vec{G}_{11}| \approx \sqrt{15.7^2 + 23.6^2} \approx 28.4 < 31.4$, so closer than $\vec{b}_2$!

Compute:

$$\vec{G}_{11} = (5\pi, -2.5\pi) + (0, 10\pi) = (5\pi, 7.5\pi) \Rightarrow |\vec{G}_{11}| = \pi\sqrt{25 + 56.25} = \pi\sqrt{81.25} = \pi \cdot 9.0139 \approx 28.32.$$

$$\vec{G}_{\bar{1}1} = (-5\pi, -2.5\pi) + (0, 10\pi) = (-5\pi, 7.5\pi), \quad \text{same mag.}$$

$$\text{Also } \vec{G}_{1\bar{1}} = (5\pi, -12.5\pi), \quad |\vec{G}_{1\bar{1}}| = \pi\sqrt{25 + 156.25} = \pi\sqrt{181.25} \approx 42.3.$$

So nearest $\vec{G}$ vectors are:

$$\pm\vec{b}_1, \quad \pm\vec{G}_{11}, \quad \pm\vec{G}_{\bar{1}1}.$$

Total of 6 neighbors $\Rightarrow$ hexagonal-like BZ.

Construct perpendicular bisectors:

- Between Γ and $\vec{b}_1 = (5\pi, -2.5\pi)$: midpoint $(2.5\pi, -1.25\pi)$, normal along $\vec{b}_1 \Rightarrow$ line: $5\pi x - 2.5\pi y = |\vec{b}_1|^2/2 = (31.25\pi^2)/2 = 15.625\pi^2$.

Divide by π : $5x - 2.5y = 15.625\pi$.

Similarly for others.

The resulting BZ is a **centrally symmetric hexagon**, slightly distorted (since $|\vec{b}_1| \neq |\vec{b}_2|$ and angle 60°).

(e) Direct-lattice planes (lines in 2D)

In 2D, Miller indices (hk) label lines. The (hk) family has normal vector proportional to $h\vec{b}_1 + k\vec{b}_2$.

But geometrically: the (hk) line intercepts axes at:

$$\vec{r} = \frac{1}{h}\vec{a}_1 \quad (k=0), \quad \frac{1}{k}\vec{a}_2 \quad (h=0).$$

General: plane (line) equation:

$$h \frac{\vec{r} \cdot \vec{b}_1}{2\pi} + k \frac{\vec{r} \cdot \vec{b}_2}{2\pi} = \text{integer}.$$

Or: $\vec{G}_{hk} \cdot \vec{r} = 2\pi n$.

Thus:

- (10): $\vec{G}_{10} = \vec{b}_1 \Rightarrow \vec{b}_1 \cdot \vec{r} = 2\pi n$. Normal to $\vec{b}_1$, i.e., direction $(-\vec{b}_{1y}, \vec{b}_{1x}) = (2.5\pi, 5\pi) \sim (1, 2)$.

So lines with slope $-b_{1x}/b_{1y} = -5\pi/(-2.5\pi) = 2$? Wait: line normal is $\vec{b}_1 = (5\pi, -2.5\pi) \Rightarrow$ line direction is perpendicular: $(2.5\pi, 5\pi) \sim (1, 2)$. So (10) lines run along $[1, 2]$ direction, spaced by $d_{10} = 2\pi/|\vec{b}_1| = 2\pi/(5\pi\sqrt{5}/2) = 4/(5\sqrt{5}) \approx 0.358$ nm.

- (11): $\vec{G}_{11} = \vec{b}_1 + \vec{b}_2 = (5\pi, 7.5\pi)$, magnitude $\pi\sqrt{81.25}$. Direction normal: $(5, 7.5) = (2, 3)$.

So lines perpendicular to $(2, 3)$, i.e., direction $(-3, 2)$. Spacing $d_{11} = 2\pi/|\vec{G}_{11}| = 2/(\sqrt{81.25}) \approx 0.222$ nm.

- (52): $\vec{G}_{52} = 5\vec{b}_1 + 2\vec{b}_2 = 5(5\pi, -2.5\pi) + 2(0, 10\pi) = (25\pi, -12.5\pi + 20\pi) = (25\pi, 7.5\pi)$.

Normal vector $(25, 7.5) = (10, 3)$. Line direction $(-3, 10)$. Very dense set.

In sketches: draw basis vectors, then families of equally spaced parallel lines; (52) is finer than (10), (11).

Exercise 4 – Structural analysis of copper

(a) Lattice constant from Bragg's law

Bragg's law:

$$2d_{hkl} \sin \theta = n\lambda.$$

Given: second-order ($n = 2$) reflection from (001) planes at $2\theta = 54.3^\circ$ $\theta = 27.15^\circ$.

For cubic lattice:

$$d_{001} = \frac{a}{\sqrt{0^2 + 0^2 + 1^2}} = a.$$

So:

$$2a \sin \theta = 2\lambda \Rightarrow a \sin \theta = \lambda.$$

Thus:

$$a = \frac{\lambda}{\sin \theta} = \frac{1.5413}{\sin(27.15^\circ)}.$$

Compute:

$$\sin(27.15^\circ) \approx \sin(27^\circ 9') \approx 0.4565.$$

More precisely: $\sin(27.15^\circ) = \sin(27.15 \cdot \pi/180) \approx \sin(0.4739 \text{ rad}) \approx 0.4566$.

Then:

$$a = \frac{1.5413}{0.4566} \approx 3.375.$$

Accepted value for Cu is 3.615, so discrepancy suggests: wait — **“(001) in fcc is not a lattice plane family”**? But the problem says “second-order Bragg reflection from the (001) planes is observed”, so perhaps they mean conventional cell planes.

But in fcc, (001) planes exist, but the ***structure factor*** kills first-order — second-order (002) is allowed.

Ah! Note: “second-order Bragg reflection from the (001) planes” means $n = 2$, $hkl = (001)$, i.e., $2d_{001} \sin \theta = 2\lambda \Rightarrow$ same as $d_{001} \sin \theta = \lambda$.

But $d_{001} = a$, so $a = \lambda/\sin \theta = 1.5413/0.4566 = 3.375$.

However, experimentally Cu has $a = 3.615$, and Cu K_α (002) reflection occurs at $2\theta \approx 50.4^\circ$.

Let's double-check: for Cu, $d_{200} = a/2 = 1.807$, Bragg: $2d \sin \theta = \lambda \Rightarrow \sin \theta = \lambda/(2d) = 1.5413/(3.614) = 0.4265 \Rightarrow \theta = 25.25^\circ$, $2\theta = 50.5^\circ$.

But given $2\theta = 54.3^\circ \Rightarrow$ our computed $a = 3.375$ is ***too small***. Possibly typo or different wavelength.

But per problem, proceed with calculation:

$$a = \frac{1.5413}{\sin(27.15^\circ)} \approx 3.375.$$

(b) Absence of first-order (001) reflection in fcc

Geometric reasoning: in fcc, atoms are at:

$$(0, 0, 0), \left(\frac{1}{2}, \frac{1}{2}, 0\right), \left(\frac{1}{2}, 0, \frac{1}{2}\right), \left(0, \frac{1}{2}, \frac{1}{2}\right).$$

The (001) planes (conventional cell) are spaced by a . Within one unit cell, there are two such planes: at $z = 0$ and $z = a/2$.

- At $z = 0$: atoms at $(0, 0, 0)$ and $(\frac{1}{2}, \frac{1}{2}, 0)$. - At $z = a/2$: atoms at $(\frac{1}{2}, 0, \frac{1}{2})$ and $(0, \frac{1}{2}, \frac{1}{2})$.

These two planes are identical and separated by $a/2$. For first-order (001) reflection, path difference between waves scattered from adjacent planes is λ . But adjacent planes are only $a/2$ apart.

The phase difference is:

$$\Delta\phi = \frac{2\pi}{\lambda} \cdot 2d \sin\theta = \frac{2\pi}{\lambda} \cdot 2 \cdot \frac{a}{2} \cdot \sin\theta = \frac{2\pi a}{\lambda} \sin\theta.$$

At Bragg angle for (001), $2a \sin\theta = \lambda \Rightarrow \sin\theta = \lambda/(2a)$. Then:

$$\Delta\phi = \frac{2\pi a}{\lambda} \cdot \frac{\lambda}{2a} = \pi.$$

So waves from the $z = 0$ and $z = a/2$ planes are ****out of phase by π ****, leading to destructive interference.

More simply: in fcc, the (001) “lattice planes” are not Bravais lattice planes — the true lattice is fcc, and the primitive translation along z is $a/2$ combined with basal shift. Thus, the first-order reflection cancels.

(c) (420) reflection in sc, bcc, fcc

The (420) reflection is allowed if the structure factor $F_{420} \neq 0$.

Structure factor for Bravais lattice with basis:

$$F_{hkl} = f \sum_j e^{2\pi i(hx_j + ky_j + lz_j)},$$

where (x_j, y_j, z_j) are basis positions in units of a .

sc: one atom at $(0, 0, 0)$. So $F = f \neq 0$ for all h, k, l .

$\Rightarrow$ **(420) allowed**.

bcc: atoms at $(0, 0, 0)$ and $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$.

$$F = f [1 + e^{i\pi(h+k+l)}] = f [1 + (-1)^{h+k+l}].$$

Nonzero iff $h + k + l$ even.

For (420): $4 + 2 + 0 = 6$ — even $\Rightarrow$ **allowed**.

fcc: atoms at $(0, 0, 0), (\frac{1}{2}, \frac{1}{2}, 0), (\frac{1}{2}, 0, \frac{1}{2}), (0, \frac{1}{2}, \frac{1}{2})$.

$$F = f [1 + e^{i\pi(h+k)} + e^{i\pi(h+l)} + e^{i\pi(k+l)}].$$

Nonzero iff h, k, l all even or all odd.

Check (420): $h = 4, k = 2, l = 0$ — all even $\Rightarrow$ **allowed**.

Geometric reasoning (as requested):

- In **sc**, every lattice point lies on the (420) planes regularly spaced — no cancellation.

- In **bcc**, the body-center atom lies exactly midway between corner atoms. The (420) planes: spacing $d_{420} = a/\sqrt{4^2 + 2^2 + 0^2} = a/\sqrt{20}$. The body-center shift is $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Dot product with $\vec{G}_{420} = \frac{2\pi}{a}(4, 2, 0)$ gives phase $2\pi(4 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2}) = 2\pi(2 + 1) = 6\pi \Rightarrow e^{i6\pi} = 1 \Rightarrow$ in-phase.

- In **fcc**, consider (420) planes. The face-centered atoms lie ***on*** the same set of planes as corners because 4,2,0 are all even — the plane equation $4x + 2y + 0z = \text{integer}$ has the same value mod 1 for $(\frac{1}{2}, \frac{1}{2}, 0)$: $4 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = 2 + 1 = 3$, integer. So all atoms lie on lattice planes $\Rightarrow$ constructive interference.

Thus, **all three lattices exhibit a (420) reflection**.